

Dictionary L2 — Region axis (Shilov vs Bulk) DERIVED geometrically + Gell-Mann–Nishijima located: the Shilov boundary’s stabilizer is $SO(4)=SU(2)_L \times SU(2)_R$ (electroweak), so leptons-on-Shilov and lepton colorlessness fall out together.

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Dictionary L2 — Region (Shilov/Bulk) + Gell-Mann–Nishijima

0. Where this sits

The keystone (#408) identified the Hall algebra’s B_2 with the $SO(5)$ of $K=SO(5) \times SO(2)$ and converted $\sigma_{BF} + \text{Charge}(SO(2)\text{-handle})$ to derived. L2 closes the Region axis (the bulk-vs-Shilov distinction Casey made central) and locates the electroweak charge structure (GMN). Region is geometric — Casey’s preferred route — and it comes out clean.

1. The Region result (RIGOROUS structure): the Shilov stabilizer is electroweak

The Shilov boundary of D_{IV}^5 is $S = (S^4 \times S^1)/Z_2$ (the Lie sphere), real dimension $(n-1)+1 = 5 = n_C$. The maximal compact $K = SO(5) \times SO(2)$ acts transitively on S , and the point-stabilizer is

$M = SO(4) = SU(2)_L \times SU(2)_R$ (dim check: $\dim K - \dim M = 11 - 6 = 5 = \dim S$ [?]).

So the ****internal symmetry** available on the Shilov boundary is $SO(4) = SU(2)_L \times SU(2)_R$ — exactly the weak-isospin group of the electroweak sector. The bulk (interior of D_{IV}^5), by contrast, carries the full $K = SO(5) \times SO(2)$.

This single fact does a lot:

	Shilov boundary	Bulk (interior)
internal symmetry	$SO(4) = SU(2)_L \times SU(2)_R$	full $SO(5) \times SO(2)$
function space	Hardy $H^2(S)$	Bergman (holomorphic L^2)
ρ -vector marker	$\rho_2 = N_c/\text{rank} = 3/2$ (Wallach point)	$\rho_1 = n_C/\text{rank} = 5/2$ (Bergman genus)
color	NONE ($SO(4)$ has no color structure)	carried by the full $SO(5)$
particles	leptons (light, fundamental, colorless $SU(2)$ doublets)	quarks / composites (colored, heavy, 3D constructions)

1.1 What this derives (two things from one fact)

- Lepton-on-Shilov (Casey’s directive): leptons live on the Shilov boundary because the boundary’s symmetry $SU(2)_L \times SU(2)_R$ is exactly the lepton (electroweak, colorless) content — leptons ARE the boundary K-types.

- Lepton colorlessness: leptons have no color because the Shilov stabilizer $SO(4)$ carries no color structure — color requires the full $SO(5)$, which is only available in the bulk. Quarks therefore must be bulk objects (colored, composite/heavy — the “3D constructions from a different region” Casey described).

So the bulk/Shilov split is not a labeling — it is the geometric distinction between the Hardy space (boundary, $SO(4)$, colorless leptons) and the Bergman space (bulk, full $SO(5)$, colored quarks), separated by the ρ -vector (3/2 vs 5/2). Region axis: DERIVED (geometric).

2. Gell-Mann–Nishijima — LOCATED (structure rigorous, coefficients pinned-not-asserted)

With weak isospin living in the Shilov stabilizer $SU(2)_L \times SU(2)_R$: - T_3 (third component of weak isospin) = the Cartan of $SU(2)_L \subset SO(4) \subset SO(5)$. - Hypercharge Y = a combination of the $SU(2)_R$ Cartan (T_{3_R}) and the $SO(2)$ charge b — the left-right-symmetric form $Y/2 = T_{3_R} + (B-L)/2$, with the $SO(2)$ supplying the $(B-L)/\text{baryon-lepton direction}$. - Gell-Mann–Nishijima: $Q = T_3 + Y/2 = T_{3_L} + T_{3_R} + (SO(2) \text{ piece})$.

So GMN’s ingredients all sit in identified Cartan directions of $K = SO(5) \times SO(2)$: T_3 in $SU(2)_L$, the rest in $SU(2)_R + SO(2)$. The STRUCTURE is rigorous (these subgroups are there, with these roles). The exact embedding coefficients (the precise linear combination giving Y , the normalization of the $SO(2)$ charge) are a genuine choice that I will PIN, not assert — consistent with the standing discipline (the genus flipped three times from memory-relabeling; I will not relabel the electroweak embedding from memory). Routed: the $SU(2)_L \times U(1)_Y \hookrightarrow SO(5) \times SO(2)$ embedding coefficients (cross-check against the derived $\sin^2 \theta_W = \text{rank}/N_c^2 = 2/9$ — the embedding must reproduce the Weinberg angle).

2.1 Consistency check available now

The derived $\sin^2 \theta_W = \text{rank}/N_c^2 = 2/9$ (Weinberg mechanism, T-series) is a constraint on the $SU(2)_L \times U(1)_Y$ embedding. Whatever the exact coefficients, they must reproduce 2/9. This is the pin’s acceptance test: the embedding is correct iff it yields $\sin^2 \theta_W = 2/9$. (So GMN is not free — it’s constrained by an already-derived quantity.)

3. The 5-tuple after L2

5-tuple axis	status after L2
σ_{BF}	DERIVED (keystone — $SO(5)$ -weight parity)
Charge	LOCATED — $SO(2) + \text{GMN structure } (T_3 \in SU(2)_L, Y \text{ from } SU(2)_R + SO(2)); \text{ exact coeffs}$ pinned-not-asserted, constrained by $\sin^2 \theta_W = 2/9$
Region	DERIVED (geometric) — Hardy(Shilov, $SO(4)$, colorless) vs Bergman(bulk, $SO(5)$, color); lepton-on-Shilov + colorlessness from one fact
Chirality	STAGED — $SU(2)_L$ vs $SU(2)_R$ asymmetry on the Shilov boundary is the handle (parity violation = the boundary picks $SU(2)_L$); L3
Winding/generation	STAGED — δ -tower; mechanism leans Coxeter/ δ ; count open
particle/antiparticle	DERIVED (engine v0.4 — E/F)

Bonus (chirality handle sharpened): chirality/parity-violation now has a concrete home — the Shilov boundary distinguishes $SU(2)_L$ from $SU(2)_R$ (the boundary dynamics can couple to one chirality). That’s the L3 lever: parity violation is the boundary breaking the $L \leftrightarrow R$ symmetry of $SO(4)$. (Stated as the handle, not yet derived.)

4. Honest scope + tier

RIGOROUS (geometry/rep theory): Shilov boundary of $D_{IV}^5 = (S^4 \times S^1)/Z_2$, dim 5; $K=SO(5) \times SO(2)$ transitive; stabilizer $M=SO(4)=SU(2)_L \times SU(2)_R$ (dim check exact); Hardy vs Bergman; ρ -vector markers (3/2 Shilov, 5/2 bulk).

DERIVED (this doc, modulo the keystone bet): Region = Hardy(Shilov) vs Bergman(bulk); lepton-on-Shilov; lepton colorlessness. These follow from the stabilizer being $SO(4)$ (colorless electroweak) — a rigorous geometric fact, given the keystone identification (K-types = particles).

LOCATED, NOT YET DERIVED: GMN exact coefficients ($SU(2)_L \times U(1)_Y \not\cong SO(5) \times SO(2)$) — structure rigorous, coefficients pinned-not-asserted, constrained by $\sin^2 \theta_W = 2/9$.

STAGED: Chirality (handle = $SU(2)_L$ vs $SU(2)_R$ boundary asymmetry = parity violation, L3); Winding (δ -tower).

Cal #27 / honesty: the Region result is genuinely clean and I'm leaning into it (it derives TWO physics facts — lepton location and colorlessness — from one geometric fact). But I'm NOT asserting the GMN coefficients (that's the exact failure mode I recorded yesterday — relabeling an embedding from memory); I route them to a pin with a built-in acceptance test ($\sin^2 \theta_W = 2/9$). And the whole thing still rides on the keystone bet (K-types = particles). Region DERIVED (geometric); Charge LOCATED; one bet upstream.

Next (my lane, L3): Chirality — parity violation as the Shilov boundary breaking the $SU(2)_L \leftrightarrow SU(2)_R$ symmetry of $SO(4)$ (the handle this doc surfaced). Then the static 5-tuple is closed except the generation/winding forcing; masses (L4) follow.

— Lyra, Dictionary L2 v0.1 (Region + GMN). GEOMETRIC RESULT: the Shilov boundary of $D_{IV}^5 = (S^4 \times S^1)/Z_2$ has point-stabilizer $M=SO(4)=SU(2)_L \times SU(2)_R$ in $K=SO(5) \times SO(2)$ (dim check exact: $11-6=5$). So Region = Hardy space (Shilov, internal symmetry $SU(2)_L \times SU(2)_R$, COLORLESS) vs Bergman space (bulk, full $SO(5)$, COLORED) — separated by the ρ -vector (3/2 vs 5/2). This DERIVES lepton-on-Shilov AND lepton colorlessness from ONE fact (the Shilov stabilizer has no color), and forces quarks to be bulk/colored/composite. Electroweak $T_3 \in SU(2)_L$ of the Shilov stabilizer; GMN structure located, exact coefficients PINNED-not-asserted (constrained by the derived $\sin^2 \theta_W = 2/9$). Region axis: DERIVED (geometric). Chirality handle surfaced: parity violation = the boundary breaking $SU(2)_L \leftrightarrow SU(2)_R$. Next: L3 chirality.